

Final Project Report Template

1. Introduction

1.1. Project overviews

To understand crop production of India from 1997 to 2021 yearly and seasonal trend, cultivated area, find the areas of high crop yield and low production to increase production

1.2. Objectives

The main objectives of this project are to study crop production patterns in India, identify areas with high and low crop yields, examine changes in agricultural production over the years, and understand the impact of seasonal variations on crop production across different regions and time periods.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Agribusiness Investor	invest in high-growth agricultural sectors	It is difficult to find out which agricultural sectors have high growth.	There is no single visual platform that shows all agricultural sectors growth.	Challenging to make a decision in which agricultural sector I should invest.
PS-2	Government Policy Analyst	Enhance agricultural productivity throughout India by analyzing crop production.	It is difficult to understand hidden patterns of crops across India.	There is no single visual platform that shows productivity pattern of crops.	Challenging to make effective policies for farmers to increase productivity and allocate resources.

2.2. Project Proposal (Proposed Solution)

Project Overview	
Objective	The main objectives of this project are to study crop production patterns in India, identify areas with high and low crop yields, examine changes in agricultural production over the years, and understand the impact of seasonal variations on crop production across different regions and time periods.
Scope	Analysis of crop production from 1997 to 2021.
Problem Statement	
Description	There is no single visual to show seasonal and yearly trends in crop production.
Impact	This visualization will help policymakers design effective strategies to improve agricultural productivity and make better use of available resources. It will also support agribusiness investors by providing insights into historical crop production, helping them identify profitable crops, emerging opportunities, and regions with steady growth potential.
Proposed Solution	
Approach	Collect the data from 1997 to 2021 on crop production across India from the ministry of agriculture and other agriculture related websites.
Key Features	Agriculture production trends from 1997 to 2021 across India.

2.3. Initial Project Planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Team members	Priority	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data collection	USN-1	Downloading Dataset	2	Goldy Gupta	High	12 August 2026	18 August 2026
Sprint-1	Data collection	USN-2	Loading the dataset	1	Goldy Gupta	Low	12 August 2026	18 August 2026
Sprint-1	Data preparation	USN-3	Clean the data	3	Goldy Gupta	High	12 August 2026	18 August 2026
Sprint-1	Data visualization	USN-4	Developing visualization	5	Goldy Gupta	Medium	12 August 2026	18 August 2026
Sprint-2	Dashboard	USN-5	Creating Dashboard	3	Goldy Gupta	High	19 August 2026	25 August 2026
Sprint-2	Story	USN-6	Creating story	3	Goldy Gupta	Medium	19 August 2026	25 August 2026
Sprint-2	Documentation and Recording	USN-7	Creating Demo	3	Goldy Gupta	High	19 August 2026	25 August 2026
Sprint 2	Documentation and Recording	USN-8	Creating Documents	5	Goldy Gupta	High	19 August 2026	25 August 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection Plan & Raw Data Sources Identification: -

Section	Description
Project Overview	To understand crop production of India from 1997 to 2021 yearly and seasonal trend, cultivated area, find the areas of high crop yield and low production to increase production
Data Collection Plan	Dataset is collected from agriculture department of India, Smart bridge site, and Directorate of Economics and Statistics.
Raw Data Sources Identified	Ministry of Agriculture and Farmers Welfare Directorate of Economics and Statistics

Raw Data Sources

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Ministry of Agriculture and Farmers Welfare	https://agriwelfare.gov.in/en/StatHortEst	CSV	1GB	Public

Dataset 2	Directorate of Economics and Statistics	https://data.desagri.gov.in/website/crops-apy-report-web	Excel	1GB	Public
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3.2. Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset 1	Some missing numbers.	Moderate	Filter the data and add missing numbers
Dataset 2	Reports related to agriculture production for each crop are separated.	High	Make a single excel file containing all reports.

3.3. Data Exploration and Preprocessing

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

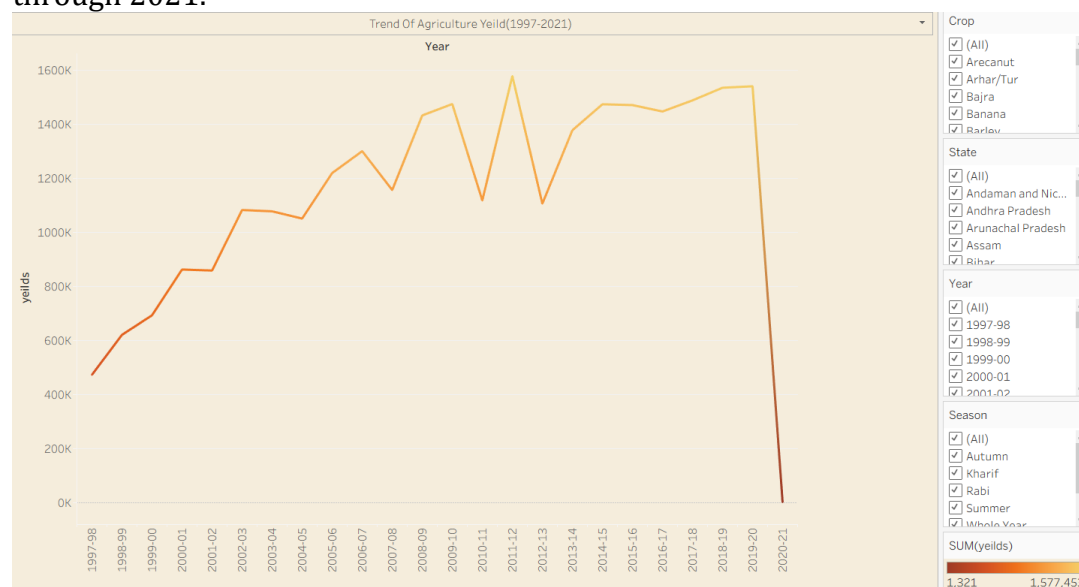
Section	Description
Data Overview	Data set is related to India's agriculture crop production yearly seasonal, state-wise.
Data Cleaning	Execute data cleaning workflows to impute missing entries, deduplicate records, and rectify data anomalies
Data Transformation	Data manipulation: filtering, sorting, pivoting, and creating calculated columns.
Data Type Conversion	Rectifying Datatype
Column Splitting and Merging	Split or merge columns as needed.
Data Modeling	Define relationships between tables and construct measures.
Save Processed Data	Save the cleaned and processed data for future use.

4. Data Visualization

4.1. Framing Business Questions

Q1. How did India's crop yield per hectare and overall farm productivity progress over the 1997–2021 period?

Ans. The line chart below illustrates the historical trajectory of agricultural yield from 1997 through 2021.



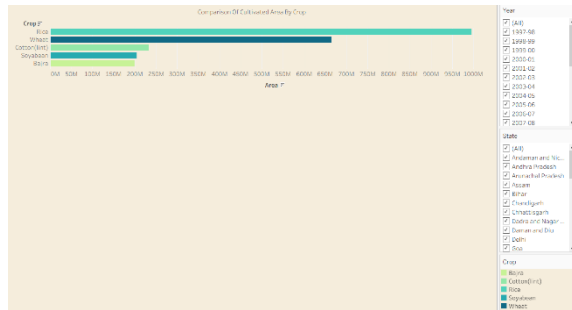
Q2. What is the total gross cropped area and aggregate agricultural yield across India?

Ans.

Indian Agriculture	
Area	4,031.13M
Production	330,997.84M
yeilds	27.43M

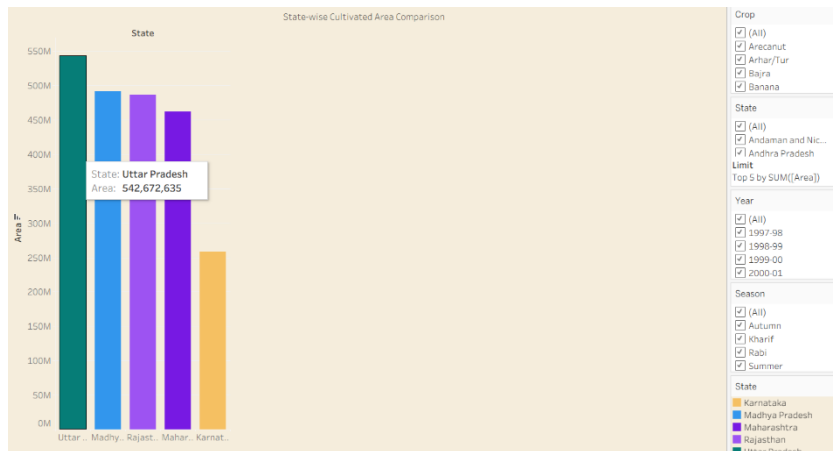
Q3. Which agricultural crops occupy the most acreage in India?

Ans. Rice has the highest cultivated area, followed by Wheat, cotton, soyabean and bajra



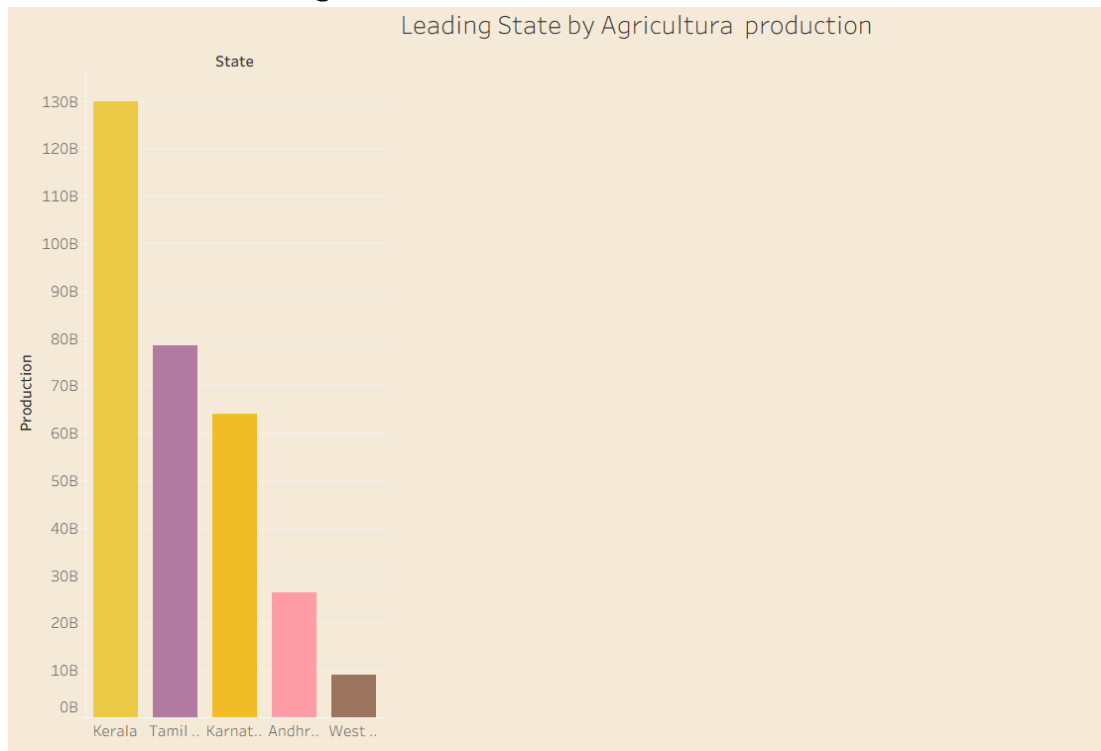
Q4. Which crops cover the largest share of India's gross cropped area?

Ans. Uttar Pradesh has the highest cultivated area followed by Madhya Pradesh, Rajasthan, Maharashtra and Karnatak.



Q5. Which Indian states lead in overall agricultural output and crop production volume?

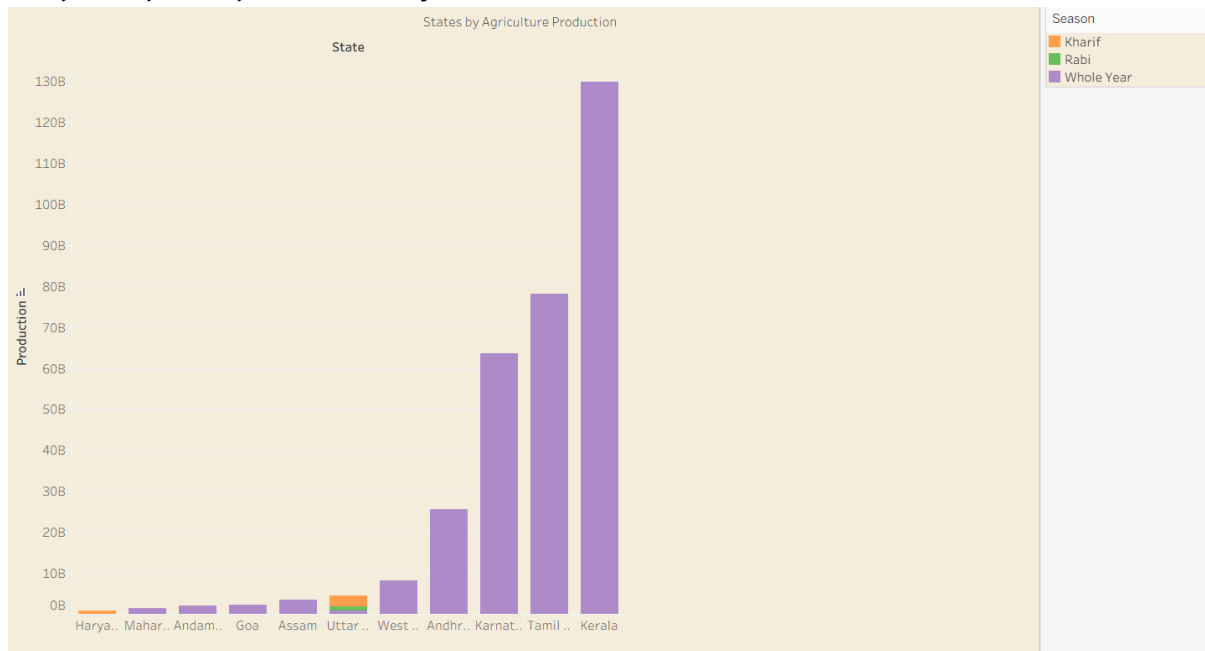
Ans. Kerala has the highest agricultural production followed by Tamil Nadu, Karnataka, Andhra Pradesh and West Bengal.



Q6. What is the bottom-ranking Indian states by cultivated land area and yield volume?

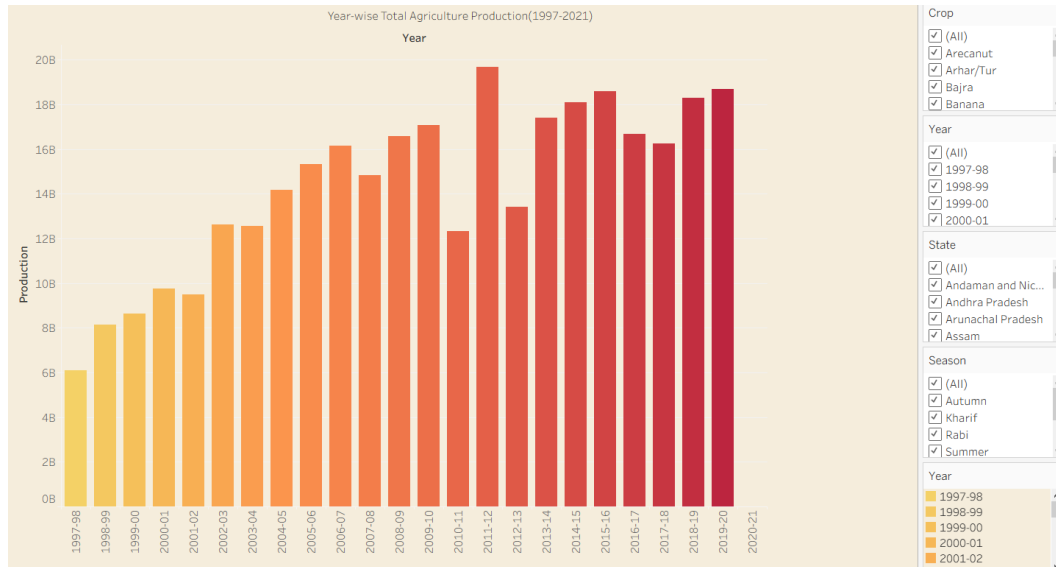
Ans. This state has low agricultural production;

Punjab,Gujrat,Rajasthan,Madhya Pradesh and follow below :-



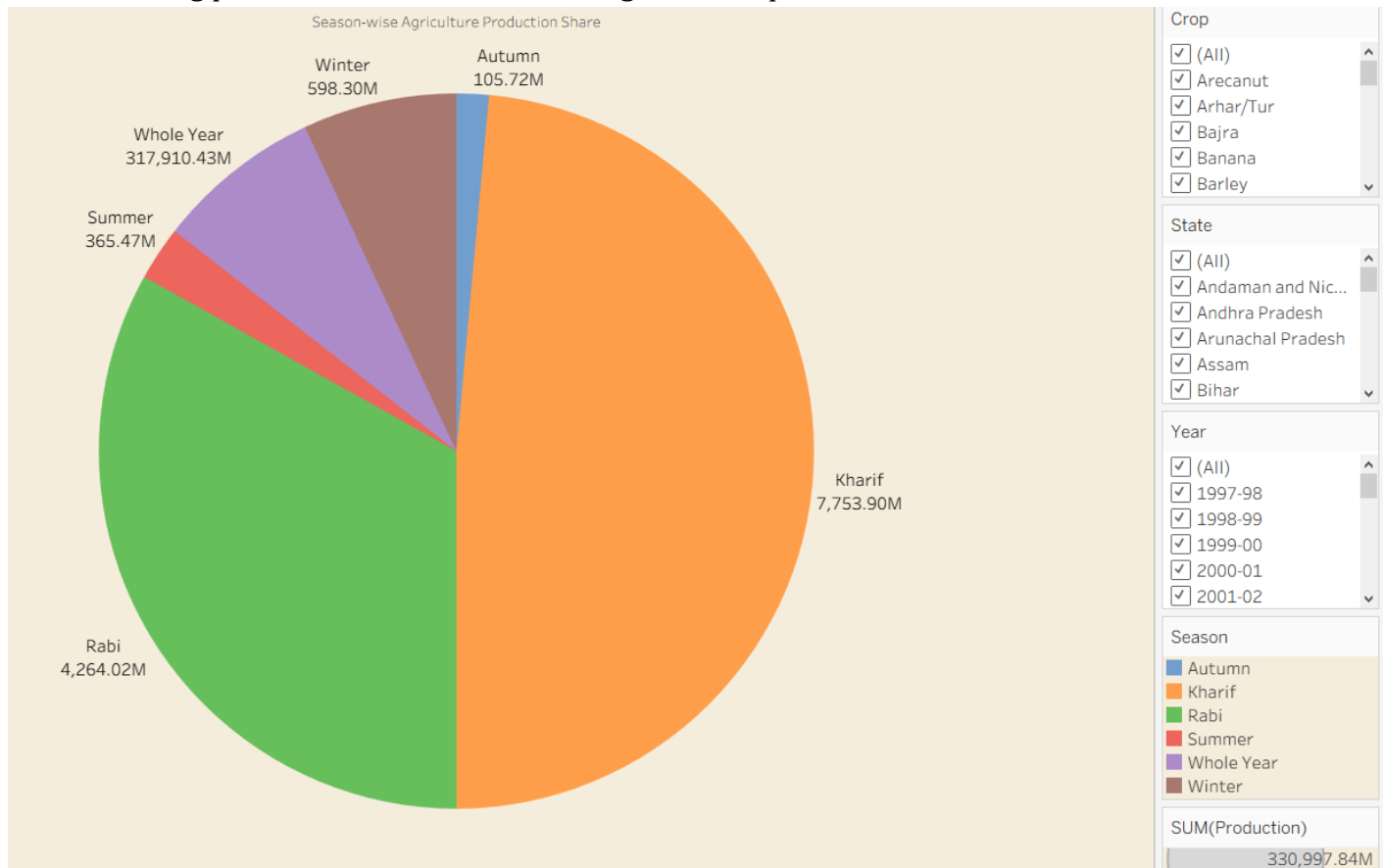
Q7. What is the year-by-year historical data for India's total crop production from 1997 through 2021?

Ans. Following bar graph shows year wise total agricultural production from 1997 to 2021: -



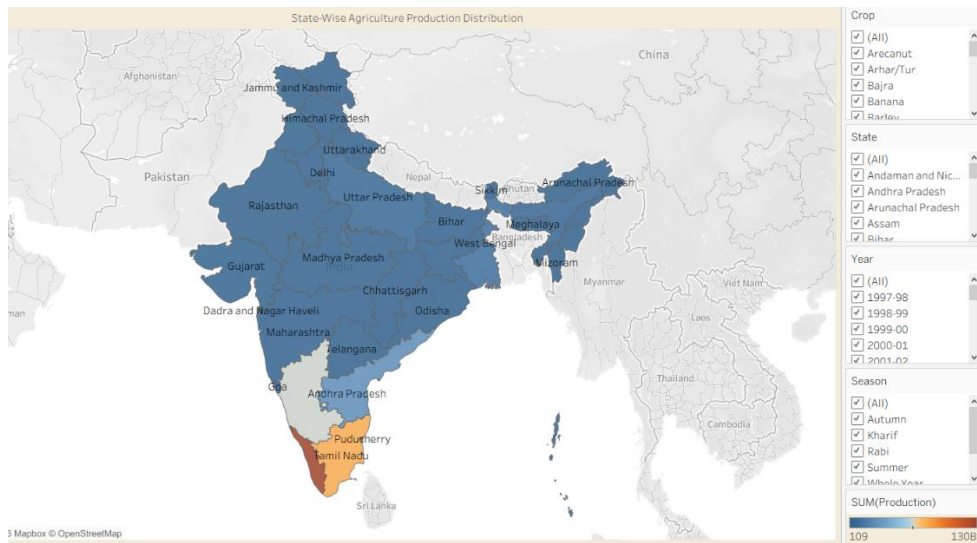
Q8. How is total agricultural production in India distributed by cropping season?

Ans. Following pie chart shows season-wise Agricultural production share: -



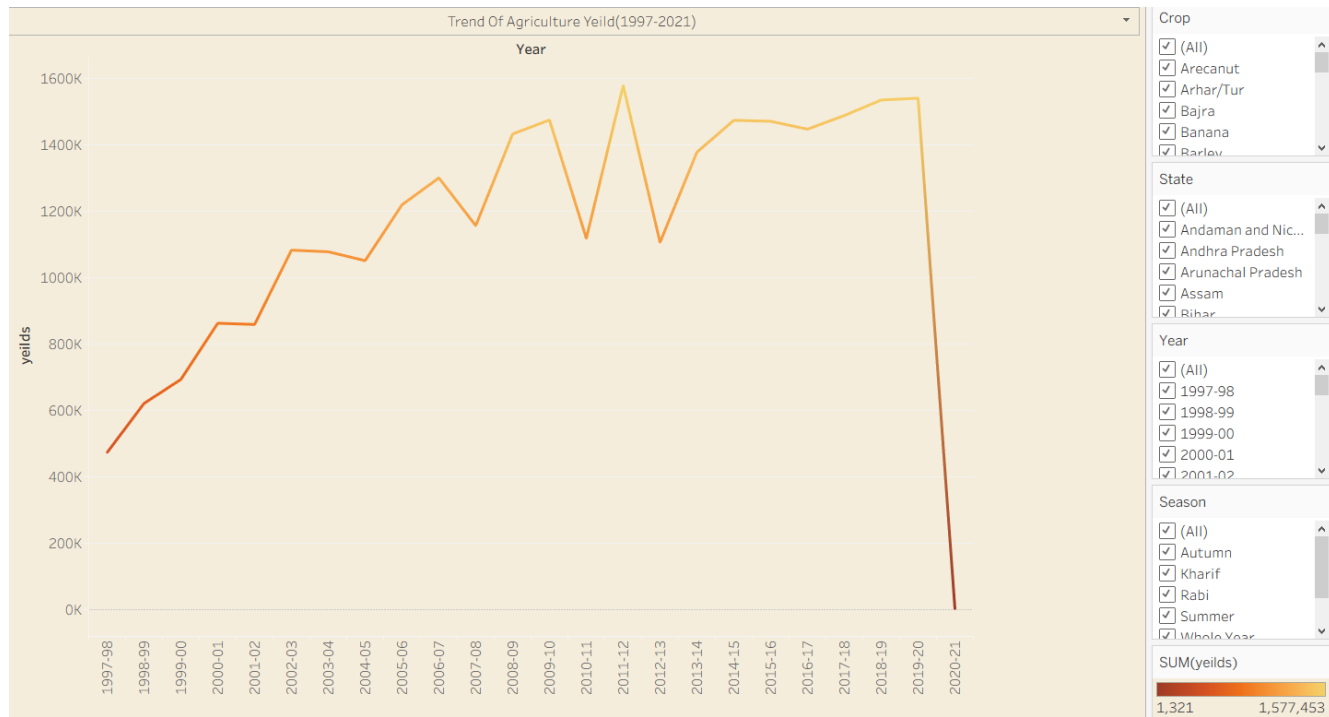
Q9. What is the relative share of each Indian state in total agricultural output?

Ans.Following map shows the states which produce during these seasons: -

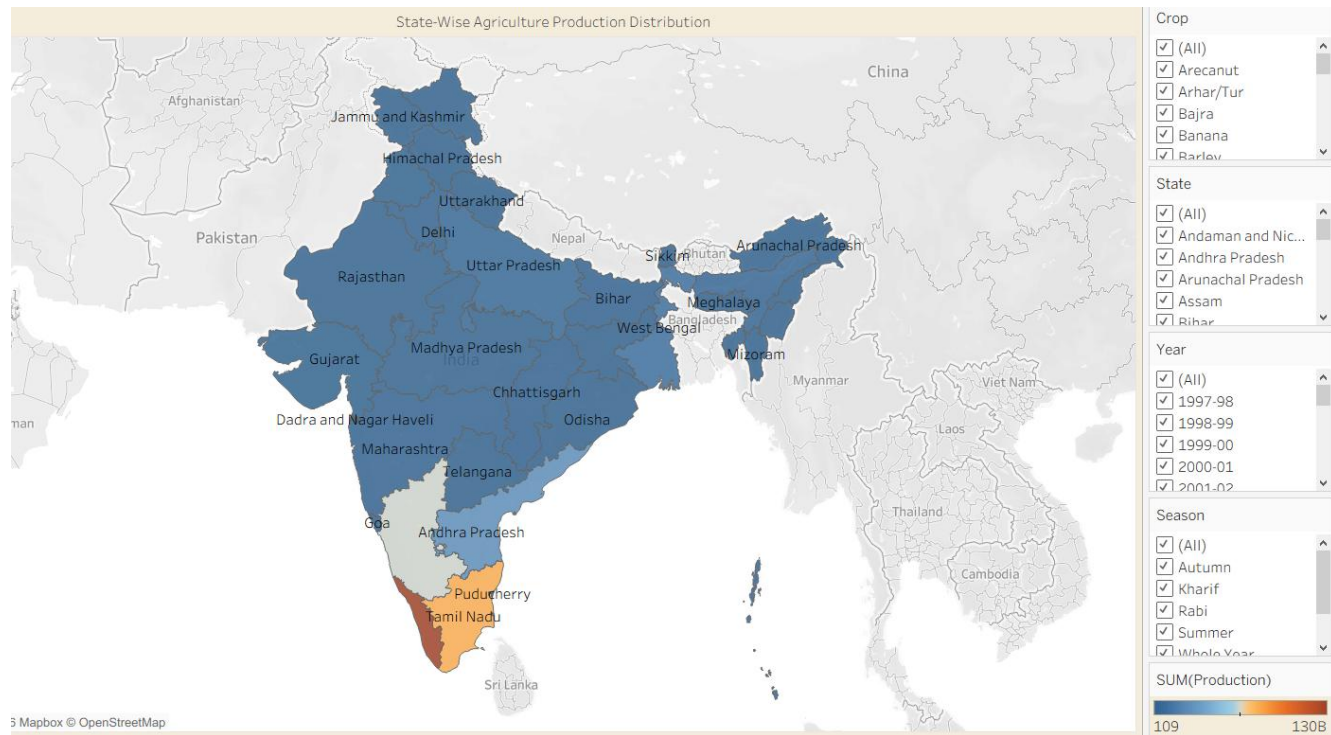


4.2. Developing Visualization

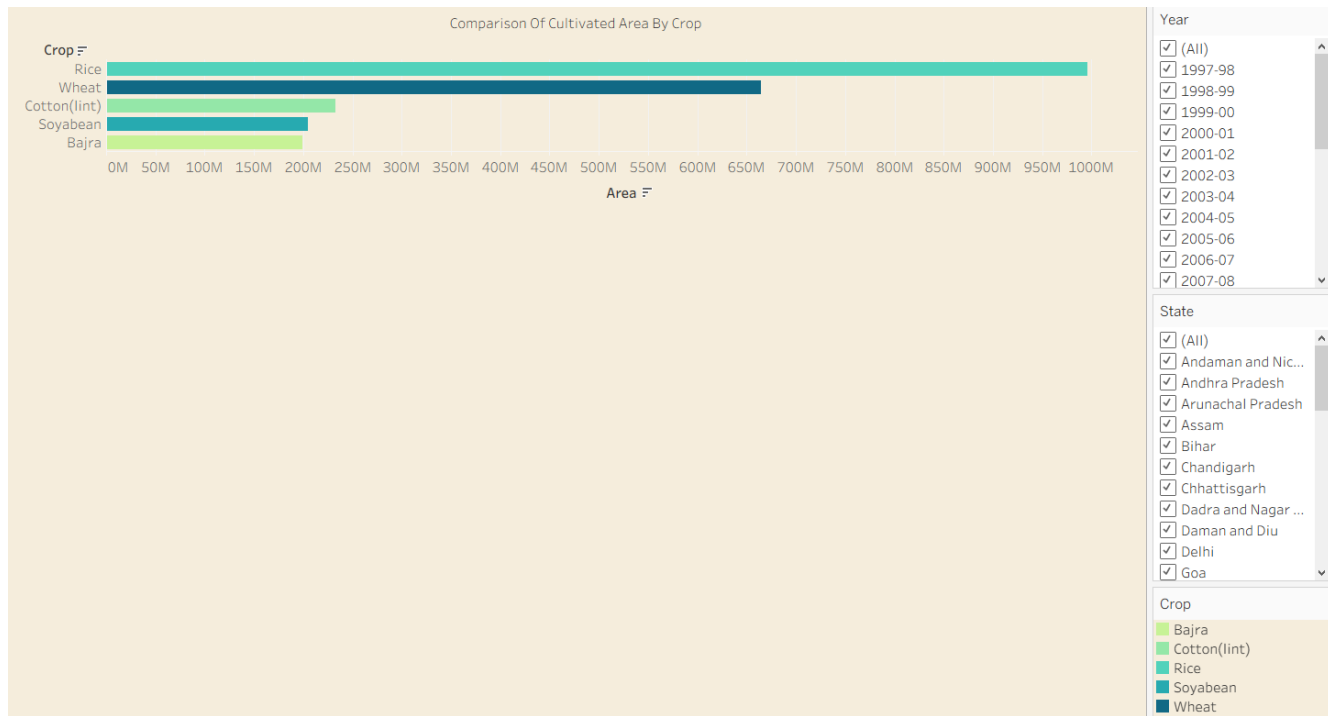
1.



2.



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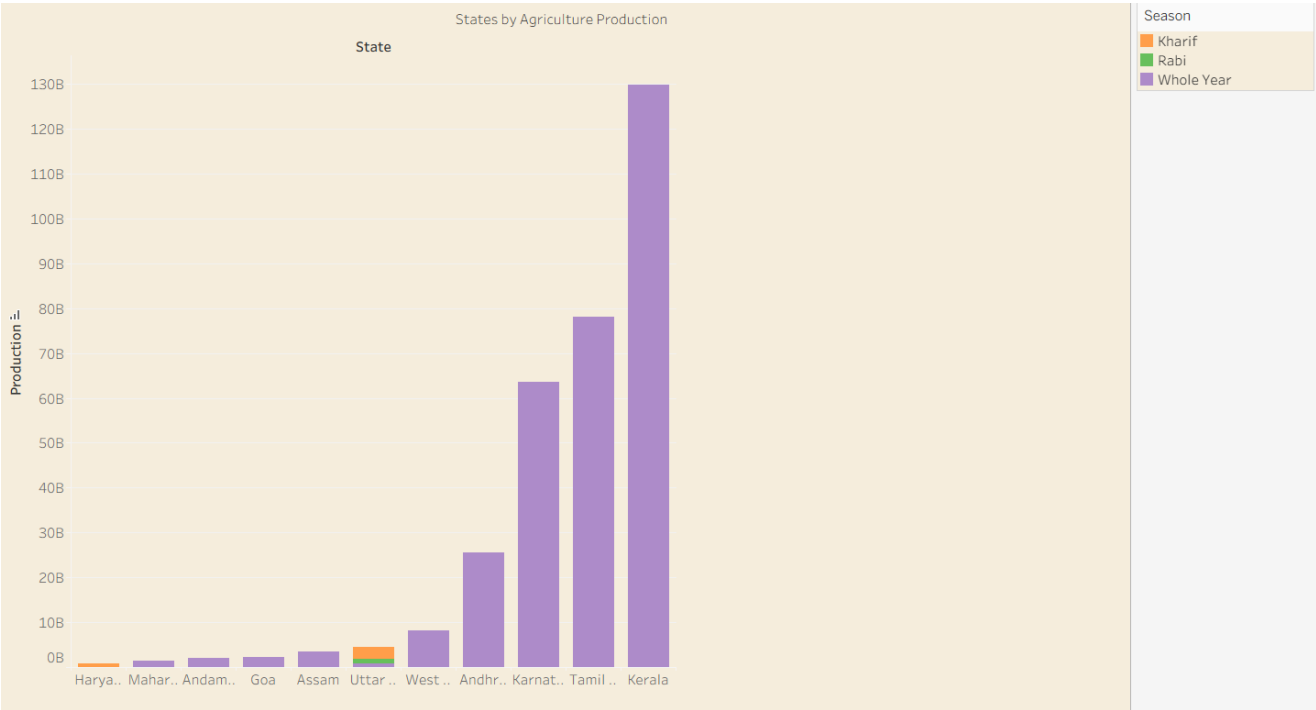
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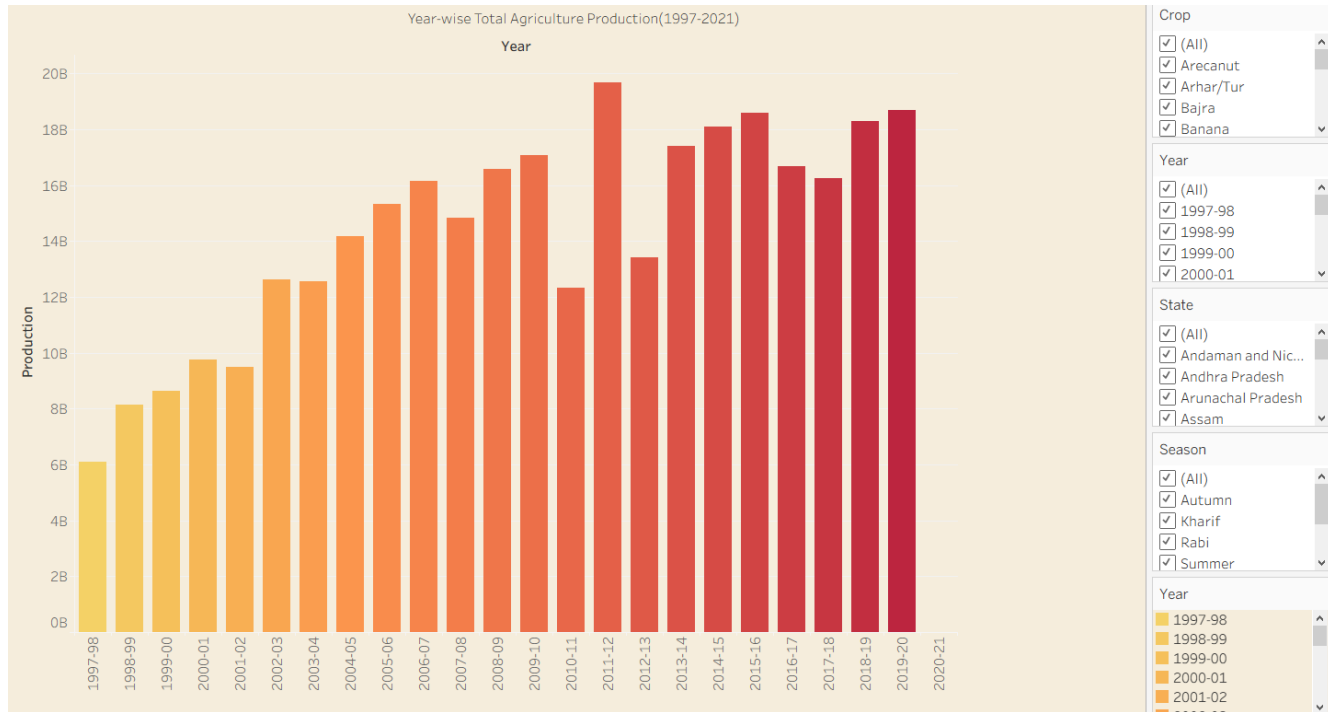
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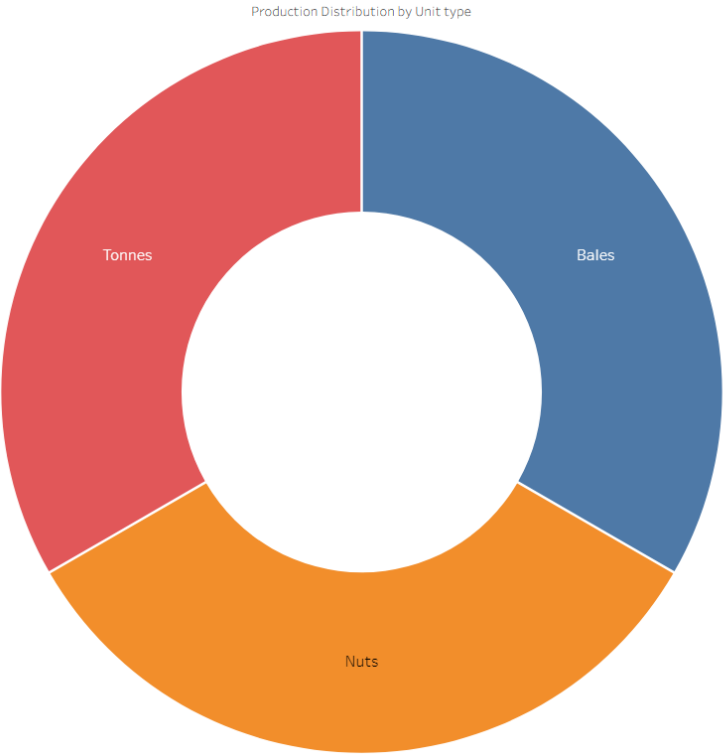
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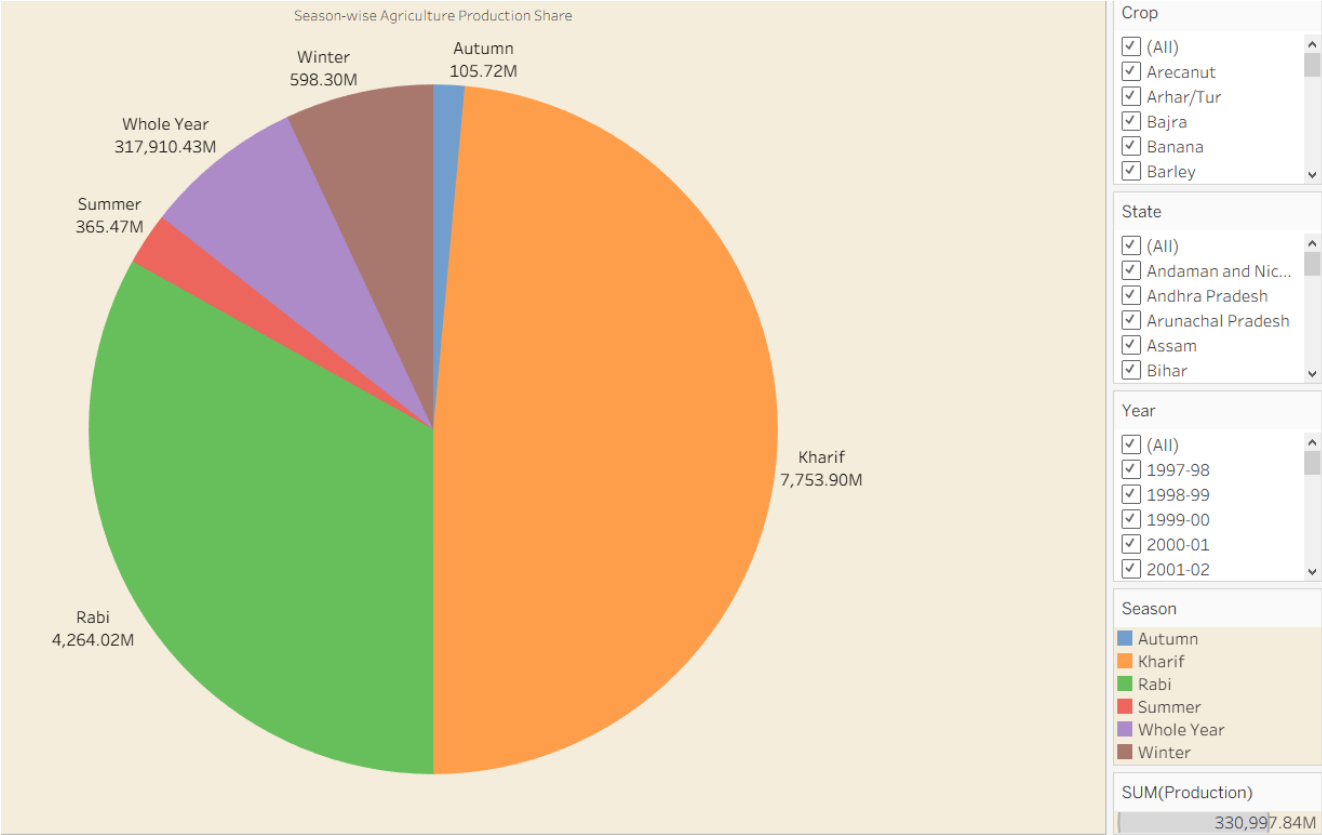


8.



Crop
☒ (All)
☒ Arecanut
☒ Arhar/Tur
☒ Bajra
☒ Banana
☒ Barley
☒ Black pepper
State
☒ (All)
☒ Andaman and Nic...
☒ Andhra Pradesh
☒ Arunachal Pradesh
☒ Assam
☒ Bihar
☒ Chandigarh
Year
☒ (All)
☒ 1997-98
☒ 1998-99
☒ 1999-00
☒ 2000-01
☒ 2001-02
☒ 2002-03
Season
☒ (All)
☒ Autumn
☒ Kharif
☒ Rabi
☒ Summer
☒ Whole Year
☒ Winter

9.



10.

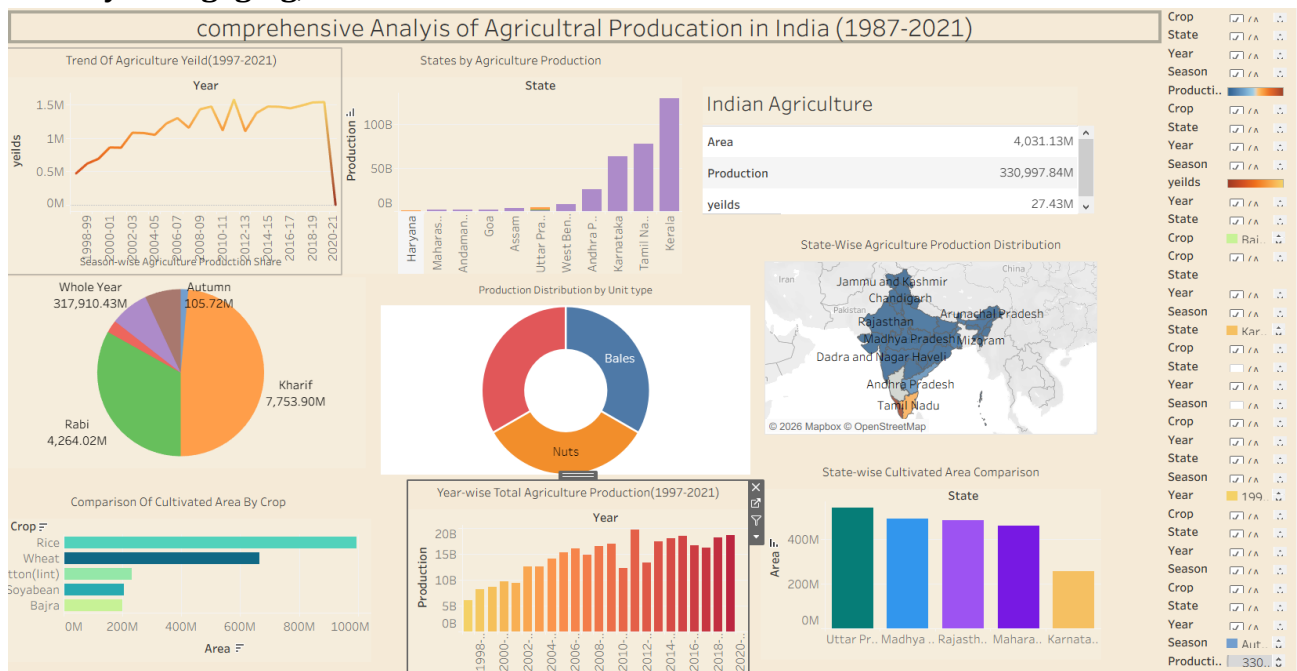
Indian Agriculture

Area	4,031.13M
Production	330,997.84M
yeilds	27.43M

5. Dashboard

5.1. Dashboard Design File

Activity 1: Engaging, interactive data dashboards.



Here are five potential outcomes from the dashboard image provided:

- **India's Agriculture performance summary: -**
 - Total area =4031.13Million Hectare
 - Total production=330,997.84 million tonnes
 - Total yield= 27.43 million tonnes
- **Leading Agricultural production State in India: -**
 - Kerala
 - Tamil Nadu
 - Karnataka
- **State with low production: -**
 - Punjab
 - Gujrat
 - Rajasthan
- **Cultivated area by crops: -**
 - Rice has highest cultivated area followed by wheat, sugarcane, coconut and potato
- **Trend of agriculture from 1997 to 2021**
 - As shown in line graph India's agriculture yield increase from 1997 to 2019 but suddenly goes down in 2020
- **Season- wise agriculture production share**
 - Kharif has highest share followed by Rabi i.e. Kharif and Rabi are the season in which production is highest in a year.

Activity 2

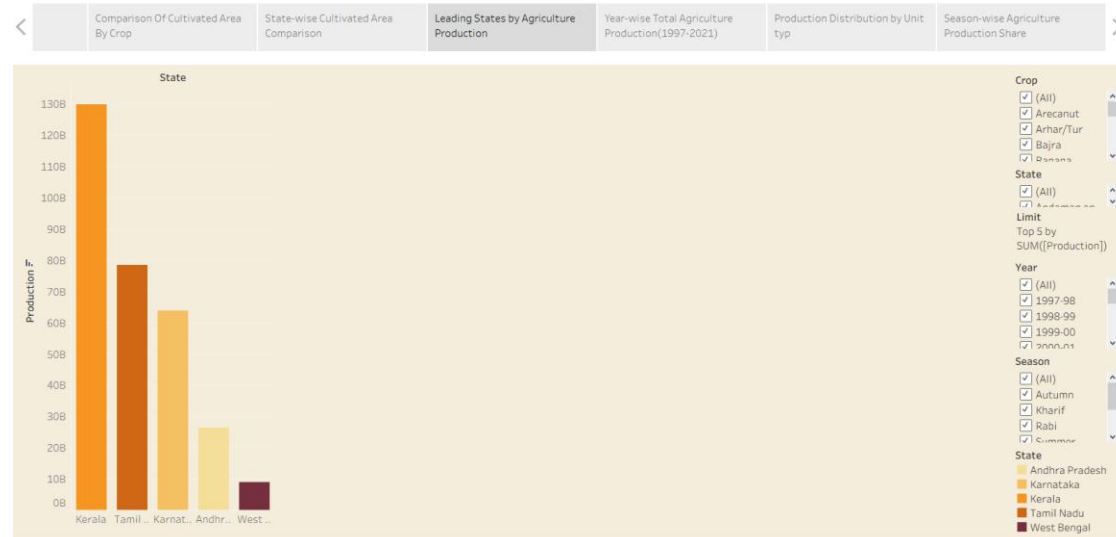
Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

https://public.tableau.com/views/goldygupta/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

6. Report

6.1. Story Design File

AGRICULTURE INSIGHT STORY



Observations:

- Yearly trend of agriculture production
- Leading states by production
- State with low production
- Seasonal production
- Comparison of States by cultivated areas
- Total area of production and yield

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/views/goldygupta/AGRICULTUREINSIGHT_STORY?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data filters

Data filters allow users to customize visual views to display specific information.

7.2 No of Visualization – 9

8. Conclusion/Observation

To conclude, agricultural output is vital for supporting livelihoods and food security across India. The data demonstrates a strong, diverse crop output driven by primary staples such as wheat and rice, while also pointing to key regional inefficiencies where high land usage yields below-average crop production.

9. Future Scope

Given that agriculture serves as a primary driver of India's economy and employment, the future of crop production remains highly favorable. Rising demographic demands and evolving dietary trends will inevitably escalate the long-term need for agricultural commodities.

10. Appendix

10.1. GitHub

<https://github.com/goldygupta9610/Project-Title-India-s-Agricultural-Crop-Production-Analysis>

Project Demo Link

https://drive.google.com/file/d/1TQXhv2AktYpCbsZCxrwwc24HFc3RvO_i/view?usp=sharing